



HIPAA-Compliant Healthcare AI Dataset Provenance on the Midnight Network using Zero-Knowledge Proofs

Hospitals prove their datasets meet HIPAA requirements — mathematically — without ever exposing a single patient record. Researchers train AI models on cryptographically verified, compliant data. Auditors verify everything on-chain.

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Healthcare AI × Midnight Network × Zero-Knowledge Proofs

Table of Contents

#	Section
1	The Problem We're Solving
2	What MediProof Does
3	Core Technology: Zero-Knowledge Proofs
4	Roles & Users
5	How It Works — Step by Step
6	Architecture
7	ZK Smart Contract
8	The Three Portals
9	Real-World Scenarios
10	Why Midnight Network?
11	Tech Stack
12	Market Opportunity

1. The Problem We're Solving

Healthcare AI is stuck. Hospitals have millions of patient records that could train life-saving AI models — for cancer diagnosis, drug response prediction, rare disease detection. But they **can't share that data**. HIPAA prohibits it.

Pain Point	Current Reality	Cost
Data sharing blocked	HIPAA prohibits record sharing	Projects never start
Drug trial delays	Insufficient training data	3–5 year timelines
AI model bias	Trained on one hospital's data	Misdiagnosis risk
Compliance overhead	Legal + de-ID pipelines per partner	\$500K–\$50M per collaboration
No cryptographic proof	Paper audit trails, easy to fake	Regulatory liability

The result:

- Drug trials take **3–5 years** partly because researchers can't access enough data
- AI diagnostic models are **biased** — trained on a single hospital's non-diverse dataset
- Compliance costs **\$500K–\$50M** per cross-institution collaboration
- Even after all that, there's **no cryptographic proof** the data was actually compliant

2. What MediProof Does

MediProof lets a hospital prove their dataset is HIPAA-compliant and that an AI model was trained on it — without revealing a single patient record.

MediProof does this by combining **Zero-Knowledge proofs** with the **Midnight Network** — a privacy-first blockchain — to create an immutable, cryptographically verifiable audit trail of healthcare data compliance, without any patient data ever leaving the hospital.

Key outcomes:

- Hospitals **monetise their data** ethically with zero liability
- Researchers **access diverse, multi-hospital datasets** for better AI models
- Auditors **verify compliance instantly** on-chain — no subpoenas, no paper trails
- FDA approvals go from **18 months** → **weeks**
- Zero patient records ever transferred — **mathematically guaranteed**

3. Core Technology: Zero-Knowledge Proofs

A Zero-Knowledge proof lets you prove a statement is **true** without revealing the underlying data.

Classic analogy: You want to prove you know a secret password, without saying the password out loud. ZK proofs make that possible — mathematically.

In MediProof:

- The hospital proves *"I have 1,200 oncology records that are de-identified"* → without sharing the records
- The researcher proves *"I trained this model on that dataset"* → without showing model weights or data
- The auditor verifies both proofs → without seeing anything private

MediProof uses **groth16** proofs compiled from **Compact** (Midnight's ZK DSL). Two circuits handle the core logic:

Circuit	What It Proves	Private Inputs	Public Output
commit_dataset	Dataset is HIPAA de-identified, ≥500 records, linked to hospital	dataset_hash, record_count, hospital_id	commitment_hash on-chain
prove_training	AI model was trained on the committed dataset	model_weights_hash, dataset_commitment, training_rows	training_proof on-chain

What goes on-chain: only the proof hashes and a timestamp.
What never leaves the hospital: patient records, raw counts, hospital identity.

4. Roles & Users

Role	Who They Are	What They Do in MediProof
■ Hospital	MGH, Mayo Clinic, any clinic with patient data	Commits a ZK proof that their dataset exists and is HIPAA-compliant
■ Researcher	MIT lab, Pfizer data science team, PhD student	Browses compliant datasets, requests AI training permission
■ Auditor	FDA, CMS, hospital compliance officer, IRB	Verifies dataset compliance and training proofs on-chain

5. How It Works — Step by Step

Step 1 — Hospital commits a dataset

- Enters: hospital name, disease category, record count, schema fields
- App generates cryptographic commitment via the **commit_dataset** ZK circuit
- Only hashes reach the Midnight blockchain — real name and count stay private
- Receives: **dataset_id** and **compliance_proof_hash**

Step 2 — Researcher finds and trains on it

- Browses published datasets (sees only: category, schema hash, ZK mode)
- Selects a dataset card and clicks 'Submit Training Request'
- 5-step ZK pipeline runs: fetch shard → privacy budget → training circuit → prove_training proof → submit to Midnight ledger
- Receives: **model_output_hash** and **training_proof_hash**

Step 3 — Auditor verifies everything

- Pastes a **dataset_id** and clicks 'Verify'
- Both proofs (compliance + training) are verified on-chain simultaneously
- Full audit log shows every event — timestamped, immutable
- No patient records, no trust required — cryptographically enforced

The entire audit trail is verifiable without a single patient record ever being shared.

6. Architecture

MediProof is a four-service system that separates concerns cleanly across frontend, backend, ZK bridge, and blockchain:

Service	Port	Language	Responsibility
Frontend	3000	React + Vite	Three-role UI: Hospital, Researcher, Auditor
Backend	8000	Python / FastAPI	REST API, hashing, DB, ZK orchestration
Midnight Service	6300	Node.js / Express	Midnight SDK bridge (WASM cannot run in Python)
Proof Server	6301	Docker	Actual ZK circuit execution (groth16)
Database	Azure	PostgreSQL 17	Stores dataset/training/audit records (hashes only)

Data flow:

```
Browser → FastAPI (Python, :8000) → asyncpg → Azure PostgreSQL (hashes only)
Browser → FastAPI → zk_client.py → Midnight Service (:6300) → Midnight SDK (WASM) →
Docker Proof Server (:6301)
Browser wallet.js → Lace extension → Midnight DApp Connector API
```

7. ZK Smart Contract

The ZK contract is written in **Compact** — Midnight Network's domain-specific language for privacy-preserving smart contracts. The compiled output (groth16 keys and ZKIR) lives in contract/dist/.

Database schema (hashes only — zero patient data):

```
datasets – dataset_id, schema_hash, disease_category, compliance_proof_hash (hospital_name
is hashed; record_count is never stored) training_requests – request_id, dataset_id,
model_output_hash, training_proof_hash audit_log – dataset_id, event_type, actor_hash,
```

```
proof_hash, timestamp
```

ZK Modes:

Mode	Condition	Behaviour
real	Contract compiled + proof server running	Actual ZK circuits via Docker proof server
mock	Proof server down or not compiled	SHA-256 hash used as proof fallback

8. The Three Portals

Hospital Portal — Commit a Dataset

The hospital declares their dataset and the app generates a cryptographic commitment — a proof that the dataset exists with specific properties, without revealing the dataset itself. Hospital name is SHA-256 hashed immediately; record count is used as a private ZK witness and never stored.

Researcher Portal — Request Training

Researchers browse all published datasets as cards (seeing only: category, schema hash, ZK mode). They select a dataset and submit a training request. A 5-step animated pipeline generates the `prove_training` ZK proof and submits to the Midnight ledger. The researcher never sees patient records, the hospital's name, or the record count.

Auditor Portal — Verify Compliance

An auditor (FDA inspector, hospital compliance officer, IRB reviewer) pastes a `dataset_id` and verifies both the compliance proof (from the hospital) and the training proof (from the researcher) simultaneously on-chain. The auto-loading audit log shows every event across all datasets — immutable and timestamped. This replaces paper audit trails, legal certifications, and data exports.

9. Real-World Scenarios

Pharma — Drug Response Model

Pfizer wants to train a chemotherapy response model across 50 hospitals in 10 countries. Without MediProof: 3–5 years of legal agreements, \$10M+ de-identification pipeline, catastrophic breach liability. With MediProof: each hospital commits a proof in minutes, Pfizer's training system generates `prove_training` proofs, FDA verifies all 50 hashes on-chain. Total time: weeks, not years. Zero patient records transferred.

Hospital — Ethical Data Monetisation

Stanford Medical has 20 years of rare disease records. Biotech companies would pay millions for access, but HIPAA creates unacceptable risk. With MediProof, Stanford publishes a ZK commitment. Companies

verify dataset properties and pay for training rights — without Stanford ever transferring data. Revenue without liability.

AI Startup — FDA Clearance

A startup has built a diagnostic AI for early Alzheimer's detection. FDA clearance requires proof the training data was HIPAA-compliant and sufficiently diverse. Currently: months of documentation and IRB approvals. With MediProof: point the FDA to on-chain proof hashes — one per hospital dataset. Approval timeline: 18 months → weeks.

Academic — Rare Disease Research

A Johns Hopkins researcher needs multi-hospital data on a disease affecting 1 in 50,000 people. 30 hospitals each commit their small datasets. Federated learning aggregates training, and MediProof generates a proof per hospital contribution. Combined model: 1,200 cases — statistically significant. Impossible without MediProof.

10. Why Midnight Network?

Midnight is a privacy-first blockchain built specifically for use cases where **data must stay private but provenance must be public**. Most blockchains (Ethereum, Solana) are fully transparent — a disaster for healthcare.

Feature	Public Chains (Ethereum/Solana)	Midnight
Transaction data	Fully public	Selectively private
ZK proofs	Add-on / complex	Native first-class
HIPAA suitability	■ Not suitable	■ Designed for it
Selective disclosure	Not supported	Built-in
Smart contracts	Solidity / Rust	Compact (ZK DSL)

11. Tech Stack

Layer	Technology
ZK Contract	Compact (Midnight Network DSL)
Blockchain	Midnight Network (Preprod testnet)
Backend	FastAPI + Python 3.14
Database	Azure PostgreSQL 17, asyncpg
Frontend	React 18 + Vite 6 + Tailwind CSS

Layer	Technology
Midnight Bridge	Node.js + Express + Midnight SDK (WASM)
ZK Proof Engine	Docker — midnightntwrk/proof-server (groth16)
Wallet	Lace browser extension (Midnight DApp Connector API)

12. Market Opportunity

Metric	Value
Global healthcare AI market	\$45B by 2026
Compliance spend (US health)	\$2.7B / year
Clinical trial data market	\$4.5B / year
ZK proof market (projected)	\$10B+ by 2030
HIPAA breach fines (avg)	\$1.5M per incident

MediProof is positioned at the intersection of healthcare compliance, AI data infrastructure, and privacy-preserving cryptography — three of the fastest-growing sectors in enterprise technology.

MediProof — Prove compliance. Protect patients. Accelerate AI.

Built on the Midnight Network · Zero-Knowledge Proofs · April 2026